Forming Algebraic Expressions

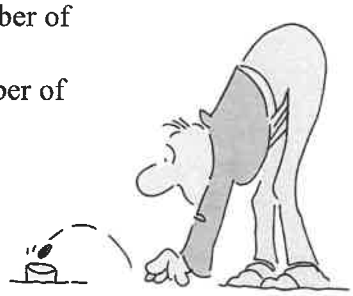
1). A gardener is planting vegetables. He plants ***m*** cabbages.
   a). He plants the same number of leeks as he does cabbages. Write an expression for the number of leeks planted.
   b). He plants 5 more carrots than cabbages. Write an expression for the number of carrots planted.
   c). He plants 2 less cauliflowers than cabbages. Write an expression for the number of cauliflowers planted.
   d). The following year he plants twice as many cabbages. Write an expression for the number of cabbages planted that year.

2). Billy, Benny and Jenny each make a tower of blocks all ***t*** blocks high.
   a). Billy adds 3 blocks to his tower. Write an expression for the number of blocks in his tower.
   b). Benny takes 6 blocks off his tower. Write an expression for the number of blocks in his tower.
   c). Jenny makes **another 2** towers of the same height. Write an expression for the number of blocks used by Jenny all together.

3). In an examination Lauren got ***x*** marks.
   a). Adam got 7 more marks than Lauren. Write an expression for the number of marks Adam got.
   b). Bob got 4 times as many marks as Lauren. Write an expression for the number of marks Bob got.
   c). Janice got half the marks that Lauren. Write an expression for the number of marks Janice got.

4). Inter-Town Rail Services introduce a new carriage for their trains. It can seat ***r*** people.
   a). On the first day of service, 4 carriages are used and each is full. Write an expression for the number of people the train is transporting.
   b). On the second day only 1 carriage is used, and this is only three quarters full. Write an expression for the number of people the train is carrying that day.

5). In a game of tiddley winks, all the counters are kept in bags each containing ***v*** counters. Alex has 5 bags, Beth has 2 bags, Colin has 7 bags and Deborah has 1 bag.
   a). Write an expression for the number of tiddley winks each person has got.
   b). Alex loses 9 counters in her games. Write an expression for the number of tiddley winks she has at the end of all her games.
   c). Beth wins 5 counters in her games. Write an expression for the number of tiddley winks she has at the end of all her games.
   d). Deborah loses half her counters in her games. Write an expression for the number of tiddley winks she has at the end of all her games.



6). A large van can hold ***g*** parcels for delivery.
   a). Quickways Ltd. have 7 of these vans. How many parcels can they deliver?
   b). Postage Force have 4 vans, but on Thursday they had to leave 6 parcels behind at the office, because all the vans were full. How many parcels were they supposed to deliver that day?
   c). Parcel Ways have 9 of these vans. On Monday the vans were not all full. They had room for another 15 parcels. How many parcels were they supposed to deliver that day?

7). A man plants 4 packets of tulips. Each packet holds f tulip bulbs. The following day he plants 9 packets of daffodils. Each of these contains g daffodil bulbs.

a). Write an expression for the number of tulips he planted.
b). Write an expression for the **total** number of bulbs he planted over the two days.

8). An egg producer designs two new boxes. The Chuckie container can hold p eggs and the Cluck Cluck container can hold q eggs.
a). Lynne buys a Cluck Cluck container. When she gets home she finds 4 eggs have been broken. How many whole eggs has she got?
b). Margaret buys 4 of the Chuckie containers. How many eggs does she buy?
c). Malcolm buys one of each container. How many eggs does he buy?
d). Javaid buys 3 Chuckie containers and 7 Cluck Cluck containers, write an expression for the total number of eggs he buys.

9). a). i). A cabbage farmer plants y seeds in a row. He plants x rows of these cabbage seeds. Write an expression for the number of cabbages he plants.
ii). The next year he plants **twice** as many seeds, write an expression for the number of cabbages he plants that year.
b). i). A carrot farmer plants f seeds in a row. He plants f rows of these carrot seeds. Write an expression for the number of carrots he plants.
ii). The next year he plants **half** as many seeds, write an expression for the number of carrots he plants that year.

10). There are two types of bus in service, the Micro-bus holding h people and the Macro-bus holding i people.
a). Hubble Buses have 16 Micro-buses and 12 Macro-buses. Write an expression for the maximum number of people they can they carry, using **all** of their buses.
b). Bubble Buses have 10 Micro-buses and 18 Macro-buses. Write an expression for the maximum number of people they can they carry, using **all** of their buses.
c). The two companies merge into one to become Hubble Bubble Buses. Write an expression for the maximum number of people this new company can carry with the combined fleet of buses.

11). On the front at Bondi Beach, two companies hire out deck chairs and sun loungers for the day. Both companies charge the same. Deck chairs cost $\$e$ to hire and sun loungers $\$f$.
a). Easy-Lie have 420 deck chairs and 136 sun loungers. Write an expression for the maximum amount of money they can take in one day.
b). Sun-Down have 340 deck chairs and 270 sun loungers. Write an expression for the maximum amount of money they can take in one day.
c). The two companies merge into one to become Sun-Lie. Write an expression for the maximum amount of money this new company can take in one day.

12). Bobby and Cilla play snap with these algebra cards. Cilla shows Bobby which pairs of cards are the same.
a). Write which cards pair up.
b). Bobby decides 10 cards are not enough for snap and writes some more. For each card below write another card that could be its's pair.

$\frac{v}{2}$ | $2v - v$ | v^2 | $2v$ | $v + 2$ | $v + v$ | $v \times v$ | $\frac{1}{2}v$ | $2 + v$ | v

i). $3v + 4v$ ii). v^3 iii). $\frac{1}{4}v$